COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:		**DATE SUBMITTED:**	
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Hands-on Activity 7.1 Managing Files and Creating Roles in Ansible

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1: Create a file and copy it to remote servers

1. Using the previous directory was created in your previous laboratory, create a directory, and name it “files.” Create a file inside that directory and name it “default_site.html.” Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.

Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Web Server</title>
</head>
<body>
<h1>Welcome to My Web Server</h1>
<p>This webpage was deployed using Ansible.</p>
<p>This file was copied from the local repository to the remote server.</p>
</body>
</html>
```

2. Edit the site.yml file and just below the web_servers play, create a new file to copy the default html file for site:

```
- name: copy default html file for site
tags: apache, apache2, httpd
copy:
src: files/default_site.html
dest: /var/www/html/index.html
owner: root
group: root
mode: '0644'
```

3. Run the playbook site.yml. Describe the changes.

None

4. Go to the remote servers (web_servers) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (default_site.html). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address. Describe the output.

The cat output of both server1 and centos1 matches the local default_site.html, confirming that the Ansible copied the file correctly to both Ubuntu and CentOS even they run different web server packages. Opening http://192.168.56.107 in a browser rendered the same page, confirming httpd was actively serving the deployed file, and same with the ubuntu.

5. Sync your local repository with GitHub and describe the changes.

```
ace@workstation:~/ansible-lab/files$ cat default_site.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, init
ial-scale=1.0">
    <title>My Web Server</title>
</head>
<body>

    <h1>Welcome to My Web Server</h1>

    <p>This webpage was deployed using Ansible.</p>
    <p>This file was copied from the local repository to th
e remote server.</p>

</body>
</html>
```

```
ace@workstation:~/ansible-lab$ cat site.yml
- hosts: web_servers
  become: true

  tasks:
    - name: copy default html file for site
      tags: apache, apache2, httpd
      copy:
        src: files/default_site.html
        dest: /var/www/html/index.html
        owner: root
        group: root
        mode: '0644'
```

```
ace@workstation:~/ansible-lab$ ansible-playbook -i inventor
y2.yaml site.yml -K
BECOME password:

PLAY [Update all servers] ********************************
*********************

TASK [Gathering Facts] ***********************************
*********************
The authenticity of host '192.168.56.104 (192.168.56.104)'
can't be established.
ED25519 key fingerprint is SHA256:8ugq9ZI2oR+rrLSBU1ZaVCRqa
LqXSkQ4oT/fxkWC38M.
This host key is known by the following other names/address
es:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
    ~/.ssh/known_hosts:6: [hashed name]
    ~/.ssh/known_hosts:13: [hashed name]
Are you sure you want to continue connecting (yes/no/[finge
rprint])? ok: [server1]
ok: [centos1]
yes
```

```
ace@workstation:~$ ssh ace@192.168.56.105
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 7.0.0-29-generic x
86_64)

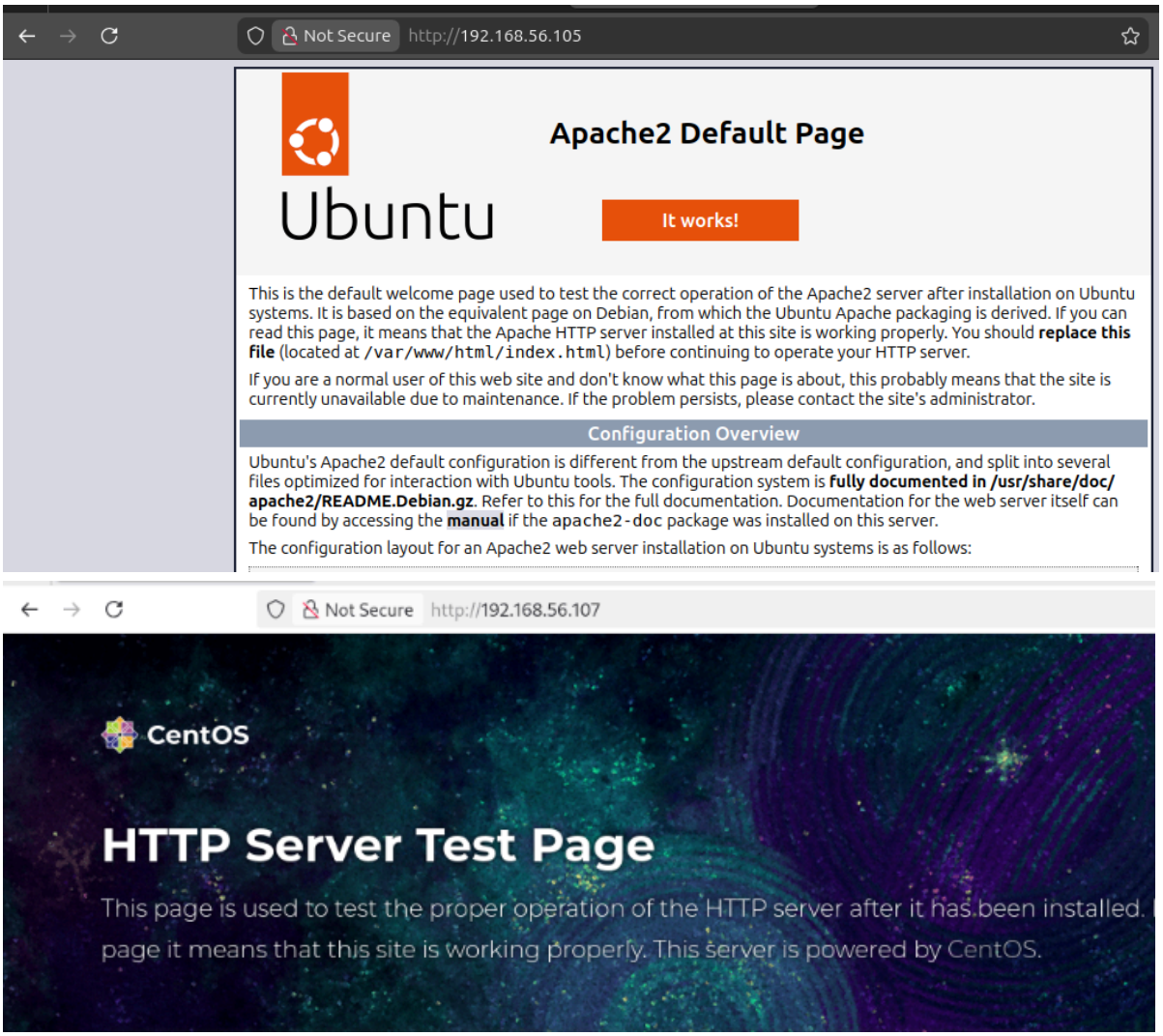
 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabl
ed.

169 updates can be applied immediately.
122 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

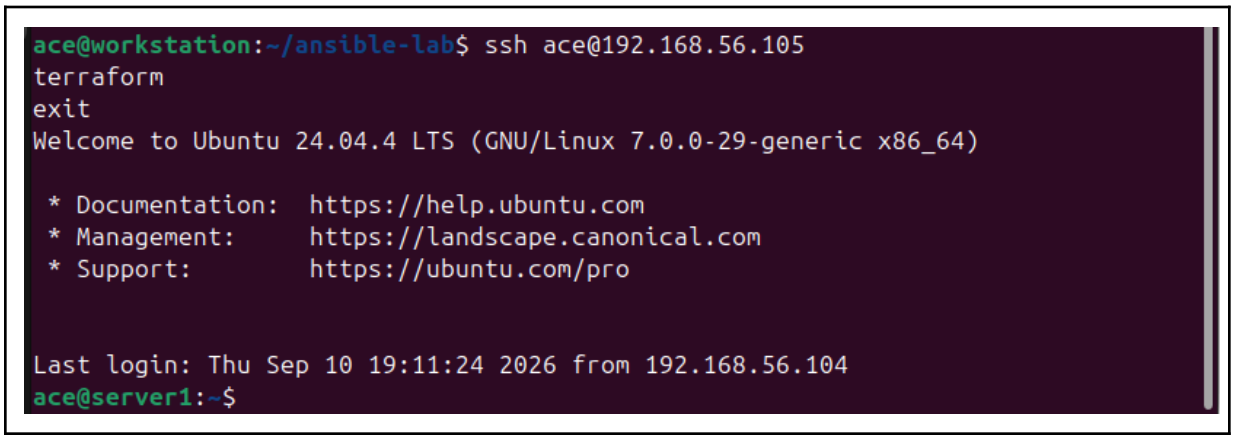
12 additional security updates can be applied with ESM Apps
.
Learn more about enabling ESM Apps service at https://ubunt
u.com/esm

Last login: Thu Sep 10 17:56:59 2026 from 192.168.56.104
ace@server1:~$ cat /var/www/html/index.html
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//
EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dt
d">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
```



Task 2: Download a file and extract it to a remote server
1. Edit the site.yml. Just before the web_servers play, create a new play:
- hosts: workstations
become: true
tasks:
- name: install unzip
package:
name: unzip
- name: install terraform
unarchive:
src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip
dest: /usr/local/bin

remote_src: yes
mode: 0755
owner: root
group: root
note: In Ansible, a play is a section of a playbook that defines which remote machines Ansible should manage and what tasks it should perform on them.
2. Edit the inventory file and add a workstation group. Add any Ubuntu remote server. Make sure to remember the IP address.
3. Run the playbook. Describe the output.
4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output.



Task 3: Create roles
1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups)
Save the file and exit.
2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers, db_servers and workstations. For each directory, create a directory and name it tasks.
3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.
4. Run the site.yml playbook and describe the output.

```
PLAY RECAP *********************************************************
**********************
centos1                    : ok=5    changed=0    unreachab
le=0    failed=0    skipped=1    rescued=0    ignored=0
server1                    : ok=5    changed=0    unreachab
le=0    failed=0    skipped=1    rescued=0    ignored=0

ace@workstation:~/ansible-lab$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What is the importance of creating roles?
   - Roles organize automation into separate, reusable components instead of a long playbook. Making projects easier to maintain and lets the same role to be reused across servers. The web_servers role deployed the HTML file to both Ubuntu and CentOS without duplicating tasks.
2. What is the importance of managing files?
   - File management modules let Ansible create and copy files across multiple servers from one control node, without manual editing. This keeps configurations consistent and efficient that saves time. This activity showed this when the same file was pushed identically to both Ubuntu and CentOS.

Conclusions/Learnings:

- Ansible can manage files and automate configuration across different operating systems from one control node. With the use of roles made the project more organized and reusable. This showed why is it efficient for server administration to relies on tools such as Ansible instead of manual setup.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Purpose of Use:

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Extent of Use:

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